PPC Gravitational Feedback Sequence

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> The relation $v^2 = v_x^2 + v_y^2 + v_z^2$ is valid in weak-field regions and in local inertial frames. In the strong-field limit, where gravitational pressure modifies the spacetime metric, velocity magnitudes should be computed using the metric of the curved spacetime.

In a curved spacetime, a more general spatial speed may involve the spatial metric components g_{ij}:

$$v^2 = g_{ij} v^i v^j.$$

PPC GRAVITATIONAL FEEDBACK SEQUENCE

Mass Density (ρ)

↓

Energy Density (E_d)

↓

Gravitational Pressure (P_g)

↓

Pressure Gradient (∇P_g)

↓

Four-Acceleration (a^μ)

↓

Momentum

↓

Energy-Pressure Tensor ($T_{\mu\nu}$)

↓

Spacetime Geometry ($G_{\mu\nu}$)

↓

Spacetime Curvature

↓

Geodesic Motion

↓

Kinetic Motion

↓

Kinetic Energy Density (K_d)

↓

Kinetic Pressure (P_{kinetic})

↓

Total Gravitational Pressure

$$P_g = P_{\text{rest}} + P_{\text{kinetic}}$$

↓

Feedback to the Gravitational Pressure Field (P_g)

Compact Form

$$\rho \to E_d \to P_g \to \nabla P_g \to a^\mu \to \text{Momentum} \to T_{\mu\nu} \to G_{\mu\nu} \to \text{Spacetime Curvature} \to \text{Geodesic Motion} \to \text{Kinetic Motion} \to K_d \to P_{\text{kinetic}} \to P_g.$$

$$P_g = P_{\text{rest}} + P_{\text{kinetic}}.$$

$$\rho \to E_d \to P_g \to \nabla P_g \to a^\mu \to \text{Momentum} \to T_{\mu\nu} \to G_{\mu\nu} \to \text{Spacetime Curvature} \to \text{Geodesic Motion} \to \text{Kinetic Motion} \to K_d \to P_{\text{kinetic}} \to P_g.$$

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